1. CHEMICAL EQUATIONS

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

1. What information do you get from a balanced chemical equation ? **9 P (AS_1)**
2. Why should we balance chemical equation ? **9 P (AS_1)**
3. Balance the following chemical equations. **9 P (AS_1) - TG-(2016)**
   a) $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$
   b) $KClO_3 \rightarrow KCl + O_2$
   c) $Hg(NO_3)_2 + KI \rightarrow HgI_2 + KNO_3$
4. Balance the following chemical equations including the physical states. **9 P (AS_1)**
   a) $C_6H_{12}O_6 \rightarrow C_2H_5OH + CO_2$
   b) $NH_3 + Cl_2 \rightarrow N_2 + NH_4Cl$
   c) $Na + H_2O \rightarrow NaOH + H_2$

APPLICATION OF CONCEPTS (AOC)

1. Balance the following chemical equation after writing the symbolic representation. **10 P (AS_1)**
   a) Calcium hydroxide$_{(aq)}$ + Nitric acid$_{(aq)}$ $\rightarrow$ Water$_{(l)}$ + Calcium nitrate$_{(aq)}$
   b) Magnesium$_{(s)}$ + Iodine$_{(s)}$ $\rightarrow$ Magnesium Iodide$_{(s)}$
2. Write the folloiwng chemical reactions by including the physical states of the substances for the following equations. **10 P (AS_1)**
   a) Sodium hydroxide reacts with Hydrochloric acid to produce Sodium chloride and water.
   b) Barium chloride and Sodium sulphate aqueous solutions react to give insoluble Barium sulphate and aqueous solution of Sodium chloride.

HIGHER ORDER THINKING QUESTIONS (HOTS)

1. 2 moles of Zinc reacts with a cupric chloride solution containing 6.023×10^{22} formula units of $CuCl_2$ calculate the moles of copper obtained. **10 P (AS_1)**

   $Zn_{(s)} + CuCl_{2\,(aq)} \rightarrow ZnCl_{2\,(aq)} + Cu_{(s)}$
2. 1 mole of propane (C_3H_8) on combustion at STP gives 'A' kilo joules of heat energy. Calculate the heat liberated when 2.4 litres of propane on combustion at STP. **10, 11 P (AS_1)**
3. Calculate the mass and volume of oxygen required at STP to convert 2.4 kg of graphite into carbon dioxide. **11 P (AS_1)**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. i) $CaCO_{3(S)} \rightarrow CaO_{(s)} + CO_{2(g)}$ ii) $2AgBr_{(s)} \rightarrow 2Ag_{(s)} + Br_{2(g)}$
Mention the types of reactions to which the above equations belong. Also mention, which of them is a photo chemical reaction. **(2016 Public)**

2. Write the equation for the reaction of zinc with hydrochloric acid and balance the equation. Find out the number of molecules of hydrogen gas produced in this reaction, when 1 mole of HC*l* completely reacts at S.T.P. **(2017 Public)**
[Gram molar volume is 22.4 litres at S.T.P., Avogadro's number is 6.023×10^{23}]

3. Write the equation for the chemical decomposition reaction of silver chloride in the presence of sunlight. **(2017 Public)**

4. How many grams of O_2 is required for combustion 480 gram of Mg ? Find the mass of 'MgO' formed in this reaction. (Mg = 24u, O = 16u). **(2019 Public)**

5. Mention the products, write balanced chemical equation when 112g. of propane (C_3H_8) is combusted. Calculate the mass of the carbon-dioxide evolved in the reaction. (Atomic mass of carbon is 12U, Atomic mass of Oxygen is 16U Atomic mass of Hydrogen is 1U). **(2022Public Model)**

6. Hydrogen reacts with Oxygen to produce water. What will be the mass of water produced if 100 grams of Hydrogen participated in the reaction? Calculate the nubmer of molecules of water produced in this reaction. [Atomic masses : H = 1u, O = 16 u] **(2022 & 2023 Public)**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

7. Keep an iron piece of solid state $CuSO_4$ crystals, does it get any reaction ? Guess the reason. **(2018 Public)**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

8. Take some water in a test tube and add concentrated H_2SO_4 to it. Shake the test tube well. If you touch the bottom of the test tube, you feel it has hot. Now, instead of H_2SO_4, if you add NaOH pellets to water in another test tube and touch the bottom of the test tube. what do you observe ? **(2016 Public)**

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Write the difference between Skelton equation and balanced chemical equation.
2. Give differences between the exothermic and endothermic reactions. **12 P**
3. $Na_2CO_3 + 2HCl \rightarrow 2NaCl \rightarrow H_2O + CO_2 \uparrow$ Identify the following
a) Reactants b) Products c) Gas d) Acid
4. Balance reaction of iron oxide with aluminium to form iron and aluminium trioxide. **15 P**
5. Balance reaction of combustion of propane. **14, 15 P**

6. What are the changes that may happen to the substances during a chemical change? Explain with an examples? **(OR)** How can you make a chemical equation more informative.

7. Calculate the volume and No. of molecules of CO_2 liberated at STP. If 50 g. of $CaCO_3$ is treated with dilute hydrochloric acid which contains 7.3 g of dissolved HCl gas. **19, 20 P**

8. $Al_{(s)} + Fe_2O_{3(s)} \rightarrow Al_2O_{3(s)} + Fe_{(s)}$ **18 P**

(atomic masses of Al = 27U, Fe = 56U, and O = 16U)

Suppose that your are asked to calculate the amount of aluminium, required to get 1120 kg of iron by the above reaction.

9. Calculate the volume, mass and number of molecules of hydrogen liberated when 230 g of sodium reacts with excess of water at STP. (atomic masses of Na = 23 U, O = 16 U, and H = 1U)

The balanced equation for the above reaction is, $2Na_{(s)} + H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$ **19 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

10. Imagine, what happend if chemical reactions were not balanced?

11. You have brushed the wall with aqueous suspension of $Ca(OH)_2$. After two days wall turned to white colour. Predict the necessary chemical reactions and equations involved in this process. **12, 13 P**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

12. What changes do you observe, when aqueous solutions of sodium sulphate is added to barium chloride solution? **17 P**

13. What changes do you observe, when aqueous solutions of Lead Nitrate is added to Potassium Nitrate solution?

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

14. Answer the following questions based on the information given in the chemical equation.

$N_2 + 3H_2 \rightarrow 2NH_3$

Atomic masses : N = 14 U

H = 1 U

i) Write the reactants and products in the above chemical reactions?

ii) How many moles of ammonia is produced?

iii) How many hydrogen atoms are present in the above chemical reaction?

iv) Find the molecular mass of NH_3 produced in the above chemical reaction.

15. Observe the given below chemical equation and answer the questions under the chemical equation

$C_3H_6 + O_2 \rightarrow CO_2 + H_2O$.

i) Write the names of the formula units form given chemical equation.

ii) Write the number of atoms of element presents in left hand side and right hand side of given chemical equation.

iii) Why the given chemical eqaution has to be balanced?

iv) Write the balanced chemical equation for given chemcial equation?

V. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

16. Write any four advantages of chemical equation.

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Potassium nitrate and Sodium Nitrate reacts separately with copper sulphate solution. Write balanced chemical equations for the above reactions. **12 P**
2. Write the difference between physical change and chemical change with suitable example.
3. Calculate the volume of O_2 at S.T.P required to completely burn 200 ml of acetylene (C_2H_2)?
4. Find the volume of the oxygen required to burn 36 g of carbon completely at STP.
5. $Ca(NO_3)_2 \rightarrow CaO + NO_2 + O_2$ Balance the adjacent equation and measure the quantity of Calcium nitrate required to get two moles of oxygen. (Ca = 40U, N=14U, O =16U)
6. Write chemical equation for the reaction between calcium carbonate and dilute hydrochloric acid and balance it. Calculate the volume and number of molecules of CO_2 liberated at STP, if 50g of calcium carbonate is treated with dilute hydrochloric acid which contains 7.3 g of dissolved HCl gas.

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

7. Why is photosythesis considered as endothermic reaction ? Give reason ? **12 P**
8. Why does respiration considered as a exothermic reaction ? Explain. **14 P**
9. What would happen if there is no method of writing chemical equations?
10. What would happen if there is no method of writing chemical equations?

III. INFORMATION SKILLS AND PROJECTS (AS_4)

11. Observe the folllowing chemical equations and answer the questions.

a) $Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)} \uparrow$

b) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + Q$

c) $N_{2(g)} + O_{2(g)} \rightarrow NO_{(g)} - Q$

d) $Fe_2O_{3(s)} + Al_{(s)} \xrightarrow{\Delta} Fe_{(s)} + Al_2O_{3(s)}$

i) Which one is exothermic reaction among all above chemical equations.
ii) In which of the above chemical equations you observed aqueous solutions. Write their names.
iii) Which is not balanced equation in the above equation.
iv) What is "Δ" represents in equation-D.

12. Answer the following questions based on the information given in chemical equation

$Fe_2O_3 + CO \xrightarrow[\text{Furnace}]{\text{In blast}} Fe + CO_2$ (Atomic masses of Fe = 56U, C=12U, O = 16U)

i) What is the coefficient of CO_2 on balancing the chemical equation
ii) Which are called reactants and products in the above chemical equation
iii) Calculate the mass of Fe(in grams) formed in the above chemical reaciton.
iv) How many grams of Fe_2O_3 is consumed in the above chemical reaction.

IV. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

13. Give the formula for quick lime. Give uses of quick lime.

2. ACIDS, BASES AND SALTS

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

1. An acid or a base is mixed with water. Is this process exothermic or endothermic one? **38 P(AS_1)**
2. Distilled water does not conduct electricity. Why ? **38 P(AS_1)**
3. Draw a neat diagram for showing acid solution conducts electricity in water. **38 P (AS_5)**
4. Why the flow of acid rain into a river make the survival of aquatic life in a river difficult? **38 P (AS_7)**

   (OR)

   What would happen if pH value of rain water is less than 5.6? **(AS_2)**
5. What is baking powder? How does it make the cake soft and spongy ? **39 P(AS_7)**

APPLICATIONS OF CONCEPTS (AOC)

1. Five solutions A, B, C, D and E when tested with universal indicator showed pH as 4, 1, 11, 7 and 9 respectively, classify the solutions as given below. **39 P(AS_1)**

   a) Neutral b) Strongly alkaline c) Strongly acidic

   d) Weakly acidic e) Weakly alkaline

   Arrange them pH increasing order of hydrogen ion concentration.
2. Why does tooth decay start when the P^H of the mouth is lower than 5.5 ? **39 P (AS_1)**

   (OR)

   What value of pH in the mouth leads to tooth decay ? Why ?
3. A milk man adds a very small amount of baking Soda to fresh milk? **40 P (AS_2)**

   a) Why does he shift the P^H of the fresh milk from acidic nature to slightly alkaline?

   b) Why does this milk take a longtime to set as curd ?
4. Plaster of paris should be stored in a moisture proof container. Explain why? **40 P (AS_1)**
5. Equal lengths of Magnesium ribbons are taken in test tubes A and B. Hydrochloric acid (HCl) is added to test tube A, while acetic acid is added to test tube B. The volume and concentration of both the acids are the same. In which test tube will the fizzing occur more vigorously and why?

   40 P (AS_1)

HIGHER ORDER THINKING QUESTIONS (HOTS)

1. Fresh milk has a pH of 6. Explain why the pH changes as it turns into curd ? **41 P (AS_3)**
2. How do you prepare an indicator using beetroot? Explain ? **41 P(AS_5)**

SUGGESTED EXPERIMENTS

1. Compounds such as alcohol and glucose contains hydrogen but are not categorized as acids. Describe an activity to prove it. **(2017 - Pub) (AS_3) 42 P**

2. What is meant by "water of crystallization" of substance ? Describe an activity to show the water of crystallization.

(OR)

Write the experimental procedure to prove the reason for colour of the copper sulphate crystals?

(OR)

Write the experimental procedure and the observations of removing water of crystallisation form the crystals of copper sulphate. **(2016 Public) (AS_3) 43P**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What is the name of the process when electricity is passed through aqueous solution of NaC*l* and what are the type of products produced. **(2015 Public) 60 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

2. What happens, if sulphuric acid (H_2SO_4) is used instead of Hydrochloric acid (HCl) in the experiment of "acid reacts with metals"? **(2023 Public Model)**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

3. List out the material for the experiment "when Hydrochloric acid reacts with $NaHCO_3$ and evolves CO_2". Write the experimental procedure. **(2018 Public) 49P**
4. Write the required materials in the experiment acids react with metals? **(2022, 2023 Public) 47P**
5. List out the materials required in the experiment of hydrogen gas is evolved when metals react with acids/ bases. Mention the precautions to be taken and experimental procedure. **(2023 Public) 47, 48 P**

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

6. Read the information given in the table and answer the following questions. **(2016 Supply)**

Solution	pH value	Reaction with Phenolphthalein solution	Reaction with Methyl orange solution
HC*l*	1	No colour change	Turns into red colour.
Distilled water	7	No colour change	No colour change
NaOH	13	Turns into pink colour	Turns into yellow colour
Lemon Juice	2.5	No colour change	Turns into red colour
NaC*l*	7	No colour change	No colour change
Baking Soda	8	Turns into pink colour	Turns into yellow colour

a) List out the acids in the above table.

b) What is the nature of the solution which gives pink colour with Phenolphthalein solution ?

c) List out the neutral solutions in the above table.

d) Name the strongest acid and the strongest base among the given solutions.

7. By observe the information given in the table and answer the questions given below the table. **(2017 Public)**

Substance (in aqueous solution)	Colour change with Blue Litmus	Colour change with Red Litmus
A	Red	No change
B	No change	Blue
C	No change	No change

i) Which one of them may be the neutral salt among A, B, C ?

ii) What may happen when some drops of phenolphthalein is added to the substance B ?

8. **Observe the table and answer the following questions.** **(2022 Public)**

S. No	Name of the substance	pH value
1.	Distilled water	7
2.	Vinegar	4.5
3.	Sodium hydroxide	13.7

i) Vinegar is added to distilled water. If methyl orange indicator is added to this solution, what will be the colour of the solution?

ii) According to the given pH values, what is the nature of the substances?

V. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

9. Is the substance present in antacid tablet acidic or basic? What is its use?**(2015, 2018 Public) 44P**

10. Why dough rises swells, when it is treated with yeast ? **(2016 Public) 39P**

11. Give two important uses for each of washing soda and baking soda.
(2017, 2019, 2022 & 2023) 58, 59 P

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What is a neutralisation reaction? Give two examples ? **44 P**

2. What is the difference between Acids and bases.

3. Explain pH scale values with respect of decrease H_3O^+ ions and increase OH^- ions.

4. Dry hydrogen chloride gas does not turn blue litmus to red whereas aqueous hydrochloric acid does. Why ? **44 P**

5. Define olfactory indicator and explain how onion and vanilla extracts used as olfactory indicators? **52 P**

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5. Define olfactory indicator and explain how onion and vanilla extracts used as olfactory indicators? **52 P**

6. How does the pH of a concentrated acid change when it is diluted ? 53 P

7. Define pH. (or) What is the relation between hydrogen ion concentration of an aqueous solution and pH? 53 P

8. How pH scale / universal indicator helped for identifying acidic, basic and neutral substances. 53 P

9. Why the milkyness is obtained when Carbon dioxide gas is passed through Lime water ? 46 P

(OR)

What changes do you observe when CO_2 gas passed through $Ca(OH)_2$ in the solution. **49 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

10. Guess and write what are the consequences, if neutralization reaction not occured?

11. Why does a Bee sting causes pain and itching? What is the remedy for it ? 55 P

12. Imagine why tooth pastes are always manufactured by mixing with mild base?

13. Why pure acetic acid does not turn blue litmus to red? Guess & write?

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

Note : For experimental questions try to practice the diagrams.

14. While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid? What is the name of the process. 46 P

15. Write an activity to find the change of colour in the reaction of an acid with a base. **(OR)** 50P

Write the experimental procedure and observations of reaction between an Acid with Bases.

16. Write an activity to show the reaction of bases with metals. 48 P

17. Write an activity to show that metal oxide react with acid is a neutralization. 45 P

(OR)

Conclude the nature of CaO by reacts with conc HCl and produce $CaCl_2$ and CO_2 gas.

18. Do acids produce ions only in aqueous solution? Prove it.

(OR)

What do acids or bases have in common. 45 P

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

19. The pH values of three solutions X, Y and Z three solutions are 13, 6, 2. Then answer the following questions ?

    i) Which solution is a strong acid ? Why ?

    ii) Which solution contains ions along with molecules of solution ?

    iii) Which solution is strong base ? Why ?

    iv) What happens to pH value when a base is added to given solution? Why it increases or decreases Give Reasons ?

20.

Salt (aqueous) solution	$NaCl$	$CuSO_4$	$NaHCO_3$	K_2SO_4
Colour with pH paper	Green	Red	Blue	Green

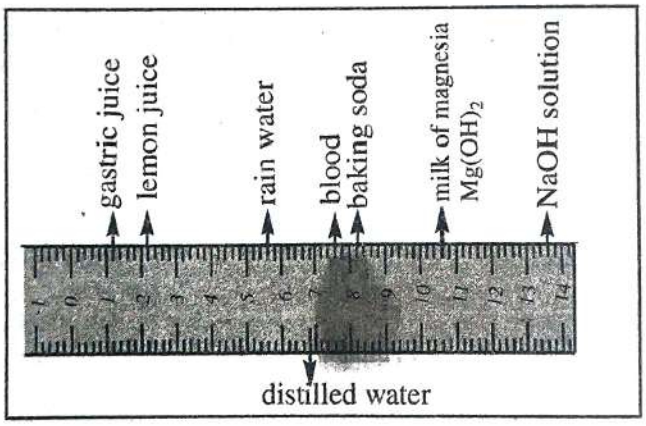
i) Why pH paper gives different range of colours, while litmus gives limited colours.
ii) In the given list acidic salt is ?
iii) What is the nature of the salt $NaHCO_3$.
iv) What is the colour of pH paper with aqueous $NaHCO_3$ solution.

21. Read the following table. Answer the questions.

Solution	Gastric fluid	Vinegar	NaOH solution	House hold bleach	Lemon juice	Baking soda	Milk of magnesia	Battery solution
pH value	1	3	13.8	13	2.2	12.8	10.6	0

i) List out acedic substances from the given table
ii) List out basic substances from the given table
iii) Name the substances used in antacids.
iv) NaOH forms common salt by reacting with ____________ ?

22.



By observing the above pH scale answer the following questions ? 56 P
i) Which of the body fluid have basic nature ?
ii) Is lemon juice a strong acid or weak acid?
iii) Which of the above liquids have strong basic character ?
iv) What is the pH of distilled water?

23.

S.No	Chemical forumula	General name	Uses
1.	$CaOCl_2$	Bleaching powder	Preparation of chloroform
2.	$NaHCO_3$	Cooking soda	Mild antiseptic
3.	Na_2CO_3	Washing soda	To remove permanent hardness of water
4.	$CaSO_4 \cdot \frac{1}{2} H_2O$	Plaster of paris	Plaster of paris making toys

Based on the above table, answer the following questions.

i) If you suffer by bitting of a honey bee, which remedy you can choose from above table to get relief?

ii) Which chemical is formed by adding of water to plaster of pairs?

iii) Write the chemical name of bleaching powder?

iv) Write the general name of chemical which is used to remove permanent hardness of water?

V. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

24. Draw the neat labelled diagram of i) pH scale **57 P** ii) warning sign figure on conc. H_2SO_4.

VI. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

25. What are the daily life applications of Calcium oxy chloride (Bleaching powder)? **58, 59 P**
26. What are the daily life applications of Calcium sulphate hemi hydrate (POP)? **59 P**
27. What is the importance of pH in your daily life. Explain with two example? **39, 54, 55 P**
28. Write the balanced equations for preparation of NaOH (Sodium hydroxide) and write their application in daily life. **59, 60 P**
29. Write any two daily life applications of salts of carbonates. **58, 59 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What is chlor-alkali process? What are the products in this process?
2. The pH of a sample of vegetable soup was found to be 6.5. How is the soup likely to taste? **54 P**
3. The dissociation of HCl in water is $HCl + H_2O \rightarrow H_3O^+ + Cl^-$ why hydrogen ions cannot exist as bare ions?

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

4. A student dipped pH paper in rain water. What would be the colour pH paper? Why?
5. Imagine and write what will happen if our stomach produces less amount of hydrochloric acid during digestion of food? **(OR)**
   What would happen if an acid is used instead of base in the perparation of tooth paste?
6. The pH of an aqueous solution decreases from 5 to 4. What will happen to the nature of the solution? **53 P**
7. Dry ammonia gas has no action on litmus paper, but a solution of ammonia in water turns red litmus blue. **44 P**
8. Why should pickles and sour substances not be kept in brass and copper vessels? **46 P**
9. What will happen if the pH value in our body increases? **54 P**
10. Why do living organism have narrow pH range? **55 P**

11. Predict the produced gas is acid or not when Sodium chloride react with Conc. sulphuric acid. **50 P**

(OR)

According to Ravi dry HC*l* gas is not an acid and dil. HC*l* is only behaves as an acid. Why? Write your opinion. **50 P**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

12. Write an activity to show that dissolving of an acid in water is an exothermic process (or) endothermic process. **51 P**

13. What changes do you observe when blue and red litmus papers are dipped in the solution formed by the reaction of HC*l* and copper oxide ? **46 P**

14. When phenolphthalein indicator is added to aqueous solution of NaOH it gives pink colour. What substance have to add for pink colour disappears.

15. Under what soil condition do you think a farmer would spread or treat the soil with quick lime (Calcium Oxide) or slaked lime (Calcium Hydroxide) or Chalk (Calcium Carbonate) **55 P**

16. In given chemical equation, the gas is evolved $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 +$ ______, Write the name of the gas is evolved and how to test the gas. **50 P**

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

17.

Salt	NaC*l*	CH_3COONa	NH_4Cl	Na_2CO_3
pH Value	7	10	5	11

From the above table. Answer the following

i) Write the salts having basic nature

ii) On basis of which acid base reaction NH_4Cl is formed?

iii) If methyl Orange is added to NaC*l* salt solution, Which colour appears?

iv) What is the use of Na_2CO_3?

18. The values from X to M is in an ascending order. Value of A is 7.

pH scale

X Y Z A K L M

Then : i) Which of the letters represents a strong acid ?

ii) Out of K,L which is a strong base ?

iii) Which of the letters represents a weak base ?

iv) Which letter represents neutral ?

3. STRUCTURE OF ATOM

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

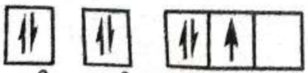
1. What information does the electronic configuration of an atom provide? **85 P (AS_1)**
2. Rainbow is an example for continuous spectrum - explain. **86 P (AS_1)**
3. How orbital is different from Bohr's orbit ? **86, 88 P (AS_1)**
4. Explain the significance of three Quantum numbers in predicting the positions of an electron in an atom. **86, 87 P (AS_1)**
5. How we are using the nl^x method in writing electronic configuration?

   (OR)

   What is nl^x method? How is it useful? **86 P (AS_1)**
6. Which electronic shell is at a higher energy level K or L ? **88 P (AS_1)**

APPLICATIONS OF CONCEPTS (AOC)

1. Answer the following questions **88 P (AS_1)**
   a) How many maximum number of electrons can be accommodated in a principal energy shell?
   b) How many maximum number of electrons can be accommodated in a sub shell ?
   c) How many maximum number of electrons can be accommodated in an orbital?
   d) How many sub shells present in a principal energy shell?
   e) How many spin orientations are possible for an electron in an orbital?
2. In an atom the number of electrons in M-shell is equal to the number of electrons in the K and L shell. Answer the following questions. **989P (AS_4)**
   a) Which is the outer most shell?
   b) How many electrons are there in its outermost shell?
   c) What is the atomic number of element ?
   d) Write the electronic configuration of the element.
3. Following orbital diagram shows the electron configuration of nitrogen atom. Which rule does not support this ? Write correct electronic configuration. N (Z = 7) **89 P (AS_1)**



$1s^2$ $2s^2$ $2p^3$

4. Write the four quantum numbers for differentiating electron of sodium (Na) atom? **89 P (AS_1)**
5. The wave length of a radio wave is 1.0 m. Find its frequency. **90 P (AS_7)**

6. i) An electron in an atom has the following set of four quantum numbers to which orbital it belong to **89 P (AS_4)**

n	l	m_l	m_s
2	0	0	+½

ii) Write the four quantum numbers for $1s^1$ electron. **90 P**

SUGGESTED PROJECTS

1. Collect the information regarding wave lengths and corresponding frequencies of three primary colors red, blue and green. **91 P (AS_4)**
2. Collect the information of historical development of the atomic theory. **90 P (AS_4)**
3. Collect the information about the scientists who developed the atomic theories. **91 P (AS_4)**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What colours are produced when iron rod is heated from low temperature to high temperature. **(2015 Public)**
2. Write the electronic configuration of the atom of an element having atomic number 11. Write the names of the rules and the laws followed by you in writing this electronic configuration. **(2017 Public)**
3. Write the symbol of the outermost shell of magnesium (Z = 12) atom. How many electrons are present in the outermost shell of magnesium ? **(2017 Public)**
4. If n = 3, mention the orbitals present in the shell and write maximum number of electrons in the shell. **(2018 Public)**
5. The electron enters into 4s orbital after filling 3p orbital but not into 3d. Explain the reason. **(2018 Public)**

   (OR)

   Explain Aufbau principle with an example. **(2024 Public) 103 P**
6. Write the main features of Bohr's model of Hydrogen atom and mention any two limitations. **(2022 Public) 97 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

7. What would be the electronic configurations of Carbon and Nitrogen , if we do not follow Hund's rule ? Guess and write. **(2022 Public)**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

8. Observe the electronic configuration given below and write number of shells, valency electrons for the given elements. **(2016 Public)**

a)

1s	2s	2p			3s
↑↓	↑↓	↑↓	↑↓	↑↓	↑

b)

1s	2s	2p			3s	3p		
↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑	↑	↑

9. Based on the information given in the table, answer the questions given below. **(2016 Public)**

S. No	Shell	K	L	M	N
1.	n	1	2	3	4
2.	l	0	0, 1	0, 1, 2	0, 1, 2, 3
3.	m_l	0	0 −1, 0, 1	0 −1, 0, 1 −2, −1, 0, 1, 2	0 −1, 0, 1 −2, −1, 0, 1, 2 −3, −2, −1, 0, 1, 2, 3

i) For the 4th main shell, how many values are there for m_l ? What are they ?

ii) For sub - shell with n = 3, l = 1, write the m_l values

iii) Write the principal quantam number value for 'N' shell. How many sub - shells are there in this main shell ?

iv) In the above table m_l and l values are given. Write a formula that gives the relationship between m_l and l based on those values.

10. The four quantum number values of the 21st electrons of scandium (Sc) are given in the following table. **(2017 Public)**

n	l	m_2	m_s
3	2	−2	$+\frac{1}{2}$

Write the values of the four quantum numbers for the 20th electron of scandium (Sc) in the form of the table.

11. Observe the information provided in the table about quantum numbers. Then answer the questions given below it. **(2017 Public)**

n	l	m_l
1	0	0
2	0 1	0 −1, 0, + 1
3	0 1 2	0 − 1, 0, +1 −2, −1, 0, +1, +2

i) Write the 'l' value and symbol of the spherical shaped sub-shell.

ii) How many values that 'm_l' takes for l = 2 ? What are they ?

iii) Write the symbols of the orbitals for l = 1 sub-shell.

iv) What is the shape of the sub-shell for l = 2 ? What is the maximum number of electrons that can occupy this sub-sell ?

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

12. Draw the shapes of 's', 'p' and 'd' orbitals. **104 P (2022 Public)**

13. Draw the shape of the all orbitals which represents the quantum numbers $n = 3, l = 2$. **(2023 Public)**

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. How do colours come from Deepavali fireworks?

2. What is Plancks quatum theory of radiation. Write an expression for it. **97 P**

3. Define the following with formula : **92 P**
   i) Wave number ii) Frequency iii) Amplitude iv) Wave length

   (OR)

   What are the factors which influence electro magnetic energy?

4. Write the set of quantum number for the electrons in a $3p_z$ orbital? **100 P**

5. Why there are exceptions in writing the electronic configurations of Chromium and Copper. **101 P**

6. Electronic configuration of Sodium is $1s^2\ 2s^2\ 2p^6\ 3s^1$. What information that it gives.

7. The outer orbital electronic configuration of an element (Z = 15) is written as the following.

   $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{}$. What is the wrong you notice. Correct it. **(OR)**

   $3s^2$ $3p^3$

   State and explain Hund's rule with an example? **102 P**

8. How do you differentiate emission and absorption spectrum? **94 P**

9. How do you differentiate band and line spectrum? **95 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

10. Imagine what happens, if quantum numbers were not discovered?

11. Predict and write, how the electron configuration of Oxygen will be changed if it is not follow Hund's rule.

12. Rani written the electronic configuration of oxygen atom as $1s^2 2s^1 2p^5$. Her friend Ravi says that electronic configuration is wrong. Do you agree with Ravi why?

13. What would be the electronic configuration of Silicon and Phosphorus, if we do not follow Hund's rule?

14. Predict the spectra produced when elements emit different flame colours **93 P**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

15. Sodium – Yellow, Copper chloride – Green, Strontium chloride – Crimson red.
Different elements produce different colours with flame test. Give the reason.

(OR)

Write an activity which shows metal produce colour in flame 92 P

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

16. If quantum numbers of an electron are

n	l	m_l	m_s
2	1	0	– 1/2

i) What is the outermost shell ?

ii) Write electronic configuration of the electron ?

iii) What is the name of the element?

iv) What is the atomic number of the element?

17. Draw the table showing relation between subshells, number of orbitals and maximum number of electrons. 101 P

Sub shells (l)	Number of orbitals ($2l+1$)	Maximum number of electrons [$2(2l+1)$]
s ($l=0$)	1	___
p ($l=1$)	___	6
d ($l=2$)	5	___
f ($l=3$)	___	14

i) What is the maximum capacity of each orbital ?

ii) How many electrons can be accommodated in the d-orbital ?

18.

Element	Electronic Configuration
C	2, 4
Na	2, 8, 1
Mg	2, 8, 2
P	2, 8, 5
Ca	2, 8, 8, 2

i) In the given list of the elements with same l value in its outer most electronic configuration.

ii) Represent electronic configuration of 2, 8, 5 in the conventional form. (nl^x method)

iii) Name the elements with same outer shell electrons.

iv) How many no. of core electrons present in 'Mg' ?

19. The Quantum number of four electrons named A, B, C, D below

Electron	n	l	m_l	m_s
A	1	0	0	+1/2
B	2	1	-1	-1/2
C	3	0	0	+1/2
D	4	1	+1	-1/2

Answer the following questions

i) Identify the electrons rotation clockwise direction

ii) Which electrons is revolving in Third orbit

iii) Which electron is very near to Atomic nucleus

iv) Which electron is far away from centre of Atom

20. A student fill the electrons into orbitals as given below.

$1s^2$ $2s^2$ $2p^4$

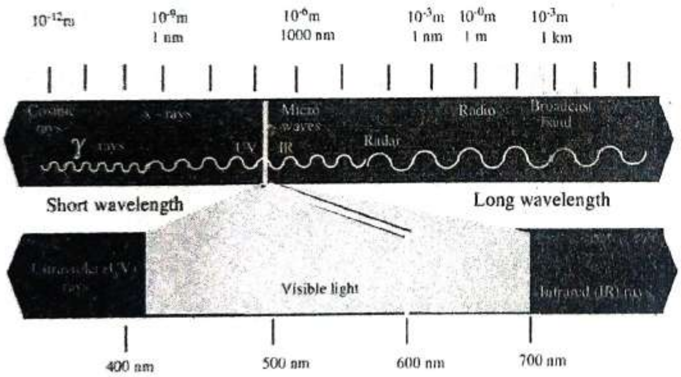
Using the above information answer the following questions.

i) Which rule is violated?

ii) Write the correct electronic configuration?

iii) What is the shape of the last orbital?

21.



Electromagnetic spectrum

By observing above electromagnetic radiation diagram answer the following questions ?

a) Name the E.M radiation with shortest wavelength

b) Name E.M radiation with highest energy

c) Name E.M radiation with used to see the nature

d) Name E.M radiation which are having Radio active nature.

V. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

22. Draw the shape of the orbitals which represents the quantum numbers n = 3, l = 2.

23. Draw the shapes of orbitals which are present in second orbit.

24. Draw the diagram (Moeller) the increasing value of n+l to write the electronic configuration. **105 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Explain the features of Bohr - Sommerfeld atomic model. **98 P**
2. What is the relation between energy and wavelength of red colour compared to blue colour? **94 P**
3. Explain briefly Heisenberg uncertainty principle? **99 P**
4. What is quantum mechanical model of atom ? Who developed it ? **96 P**
5. Explain briefly the wave nature of electron. **99 P**
6. What are the charactesistics of electromagnetic spectrum?

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

7. What are the concequences if spectroscopy was not developed?
8. What happens if the electrons do not revolve around the nucleus?

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

9. Answer the following questions from the information given below in tabular column.

S.No.	Element	n	1	2		3			4			
		l	0	0	1	0	1	2	0	1	2	3
		subshell(l)	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f
1	F		2	2	5							
2	Sc		2	2	6	2	6	1	2			
3	Ni		2	2	6	2	6	8	2			

i) What is the valency of Fluorine?

ii) Mention the number of protons in the Scandium.

iii) Write the maximum number of electrons can be accommodated in d - orbitals

iv) To which block in the periodic table does Nickel belongs?

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

10. Draw the neat labelled diagrams of the following :

i) Electro magnetic spectrum (EMS). **96 P**

ii) Electro magnetic wave (EMW). **95 P**

4. CLASSIFICATION OF ELEMENTS - THE PERIODIC TABLE

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

1. What are the limitations of Mendeleeff's periodic table? How could the modern periodic table overcome the limitations of Mendeleeff's table ? **129 P (AS_1)**

2. Define the modern periodic law. Discuss the construction of the long form of the periodic table. **130 P (AS_1)**

3. Explain how the elements are classified into s, p, d and f-block elements in the periodic table and give the advantage of this kind of classification. **130, 131 P (AS_1)**

4. Write down the characteristics of the elements having atomic number 17. **132 P (AS_1)**

   i) Electronic configuration ______

   ii) Period number ______

   iii) Group number ______

   iv) Element family ______

   v) No. of valence electrons ______

   vi) Valency ______

   vii) Metal or non-metal ______

5. Complete the following table using the periodic table. **132 P (AS_1)**

Period number	Filling up orbital's (sub shells)	Maximum number of electrons, filled in all the sub shells	Total no. of elements in the period
1			
2			
3			
4	4s, 3d, 4p	18	18
5			
6			
7	7s, 5f, 6d, 7p	32	incomplete

6. Complete the following table using the periodic table. **132 P (AS_1)**

Period number	Total no. of elements	Elements		Total no. of elements in			
		From	To	s-block	p-block	d-block	f-block
1.							
2.							
3.							
4.							
5.							
6.							
7.							

7. Why was the basis of classification of elements changed from the atomic mass to the atomic number. **144 P (AS_1)**

8. What is a periodic property ? How do the following properties change in a group and period explain. (Periodic trends) **142 P (AS_1)**

   i) A) Atomic radius B) Ionization energy
   C) Electron affinity D) Electronegativity (oxidising agent)

   ii) Explain the ionization energy order in the following sets of elements :
   A) Na, Al , Cl B) Li, Be, B C) C, N, O
   D) F, Ne, Na E) Be, Mg, Ca

9. What is ionisation potential? Explain factors affecting ionisation potential. **152 P (AS_1)**

10. Comment on position of Hydrogen in periodic table. **145 P (AS_1)**

APPLICATIONS OF CONCEPTS (AOC)

1. Given below is the electronic configuration of elements A, B, C, D. **133 P (AS_1)**
   A) $1s^2\, 2s^2$ B) $1s^2\, 2s^2\, 2p^6\, 3s^2$ C) $1s^2\, 2s^2\, 2p^6\, 3s^2\, 3p^3$ D) $1s^2\, 2s^2\, 2p^6$
   i) Which are the elements coming with in the same period ?
   ii) Which are the ones coming with in the same group ?
   iii) Which are the noble gas elements ?
   iv) To which group and period does the elements 'C' belong

2. s - block and p - block elements (except 18^{th} group elements) are sometimes called as 'Representative elements' based on their abundant availability in the nature. Is it justified ? Why ? **133 P (AS_1)**

3. The electronic configuration of the elements X, Y and Z are given below. **133 P (AS_4)**

A) X = 2 B) Y = 2, 6 C) Z = 2, 8, 2

i) Which element belongs to second period ?

ii) Which element belongs to 2nd group ?

iii) Which element belongs to 18th group?

iv) What is the valency of element Y?

4. a) State the number of valence electrons, the group number and the period number of each element. Given in the following table. **133 P (AS_1)**

Element	Valence electrons	Group number	Period number
Sulphur			
Oxygen			
Magnesium			
Hydrogen			
Fluorine			
Aluminium			

b) State whether the following elements are in a same Group (G), Period (P) or Neither (N). Group nor Period indicating the letters G (or) P or N. **134 P (AS_1)**

Element	Group / Period / Neither group nor period
Li, C, O	
Mg, Ca, Ba	
Br, C*l*, F	
C, S, Br	
A*l*, Si, C*l*	
Li, Na, K	
C, N, O	
K, Ca, Br	

5. Identify the element that has the larger atomic radius in each pair of the following and mark it with a symbol (✓). **134 P (AS_1)**

i) Mg or Ca ii) Li or Cs iii) N or P iv) B or A*l*

6. Identify the element that has the lower Ionization energy in each pair of the following and mark it with a symbol (✓). **134 P (AS_1)**

i) Mg or Na ii) Li or O iii) Br or F iv) K or Br

7. How does metallic character change when we move ? (Electropositivity, reducing agent)
i) Down a group ii) Across a period **134 P (AS_1)**

8. Name two elements that you would expect to have chemical properties similar to Mg. What is the basis for your choice ? **146 P (AS_2)**

9. On the basis of atomic numbers predict to which block the elements with atomic number 9, 37, 46 and 64 belongs to ? **135 P (AS_2)**

10. Using the periodic table, predict the formula of compound formed between the element X of group 13 and another element Y of group 16. **135 P (AS_2)**

11. An element X belongs to 3rd period and group 2 of the periodic table. State **139 P (AS_2)**
    i) The number of valence electrons in it
    ii) The valency
    iii) Whether it is metal or a nonmetal

12. An element has atomic number 19. Where would you expect this element in the periodic table and why? **135 P (AS_2)**

HIGHER ORDER THINKING QUESTIONS (HOTS)

1. How the positions of elements in the periodic table help you to predict its chemical properties? Explain with an example. **133 P (AS_7)**

2. In period 2, element X is to the right of element Y. Then, find which of the element have. **136 P(AS_1)**
    i) Low nuclear charge
    ii) Low atomic size
    iii) High ionization energy
    iv) High electronegativity
    v) More metallic character

EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

1. Aluminium does not react with water at room temperature but reacts with both dil. HCl and NaOH solutions. Verify these statements experimentally. Write your observations with chemical equations. From these observations, can we conclude that Al is a metalloid ? **136 P**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. How are the elements arranged into groups and periods in the Modern Periodic Table ? Elements in a group possess similar properties, but elements in a period do not show similarities in their properties. Why ? **146 P (2017 Public)**

2. What is electron affinity? What are the factors influence electron affinity? **150 P (2017 Public)**

3. Explain how does the atomic size and ionization energy changes in groups and periods in the Modern Periodic Table. **(2022 Public, 2023 supply)**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

4. Which one between Na and Na^+ would have more size ? Why ? **148 P**

(OR)

Between a neutral atom and its cations which has bigger size ? Why ? **(2016 Public)**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

5. Two elements X and Y belong to groups 1 and 2 respectively in the same period of the Periodic Table. Compare these elements with respect to : **(2015 Public)**
   i) number of electrons in their outer - most orbit.
   ii) their atomic size and their valencies.
   iii) their ionisation energy and metallic character.
   iv) formulae of their chlorides and sulphates.

6. In the table given below, names of some elements of families are given . Based on this, fill the information in the empty boxes. **(2015 Supply)**

S.No.	Name of the Elements family	Elements From	Elements To	Valency shell Configuration	Valency electrons	Valency
1.	Alkali metal family	Li	Fr			
2.	Alkali earth metal family	Be	Ra			
3.	Carbon family	C	Fl			
4.	Halogen family	F	At			

7. Observe the information provided in the table and answer the questions given below it.

Element	Na	C	Ca	P	Ti	Ni
Number	11	6	20	15	22	28

   i) What are the s-block elements in the table ?
   ii) What are the 'p' block and 'd' block elements in the table ? **(2017 Public)**

8. Observe the given table and answer the following question. **(2019 Public)**

S.No.	Electron configuration
1.	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
2.	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$
3.	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

   i) Mention the divalent element name.
   ii) Name the element belongs to 3rd period and VA group.

9. Observe the given table and answer the following questions. **(2023 Public)**

Atomic Number	11	12	13	14	15	16	17
Name of the Elements	Na	Mg	A*l*	Si	P	S	C*l*

   i) Which period does the elements belongs ?
   ii) Mention the non- metals in the above.
   iii) Which element has more atomic radius in the above ?
   iv) From left to right how is the metallic character changes ?

10. Answer the following question by using the above data (2022 Public)

2^{nd} period elements	Li	Be	B	C	N	O	F
Atomic number	3	4	5	6	7	8	9
Atomic radius (in pm)	152	111	88	72	74	66	64
Electro negativity (in ev).	1.0	1.47	2.0	2.5	3.0	3.5	4.0

i) How does the capacity of loosing electrons changes in the 2^{nd} period from left to right.

ii) How does atomic size changes in the 2^{nd} period?

iii) Mention the valance shell of the elements from Li to F.

iv) Mention the position of the element 'N' in the periodic table.

11. Observe the information given in the table. Answer the questions given below : (2024 Public)

Element	Electronic configuration
Be	$1s^2$ $2s^2$
Mg	$1s^2$ $2s^2$ $2p^6$ $3s^3$
P	$1s^2$ $2s^2$ $2p^6$ 3^2 $3p^3$
Ne	$1s^2$ $2s^2$ $2p^6$

i) Which of the given are s - block elements?
ii) Which of the given has least valency?
iii) Which of the given is 15^{th} Group element?
iv) Which of the given are of same Period?

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What are noble gases? What is the general electronic configuration of noble gases? **141, 142 P**
2. Define the following :
   i) Mendeleef's periodic law **138 P** ii) Moseley's periodic Law.
3. Define Dobereiners law of triads. Give examples of two Dobereiner triads. **138 P**
4. What is electronegativity? What are various methods used to determine electronegativity ? Explain. **147 P**
5. Define New land's concept of octaves? What were the limitations of Newland's law of octaves ? **138 P**
6. Why nitrogen has less electron affinity value compared to oxygen? **146 P**
7. Newlands proposed the law of octaves. Mendeleeff suggested eight groups for elements in his table. How do you explain these observations in terms of modern periodic classification? **139 P**
8. The calculated electron gain enthalpy values for alkaline earth metals and noble gases are positive. How can you explain this ? **148 P**

9. Second ionization energy of an element is higher than its first ionisation energy. Why? **149 P**

(OR)

Why is it difficult to remove an electron from Mg^+ ion compared to neutral Mg atom?

10. Ramya says Silicon is a metal. Sudha says Silicon is a non metal . Whom do you support explain with suitable examples ?

11. **3rd period :** Na, Mg, A*l*, Si, P, S, C*l* **4th group** : C, Si, Ge Sn Pb

Based on above information classify elements into metals, metalloids and non metals

12. Define valency and how valencies will be varied for different elements upto group no. - IV A, and above group no. IVA. **144, 145, 152 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

13. Which one between C*l* and Cl^- would have more size ? Why ? **148 P**

(OR)

Between a neutral atom and its anion which has bigger size ? Why ?

14. Imagine and write, if all the elements are not arranged in a particular table. What could be the consequences?

15. The second period element 'F' has less electron gain enthalpy than the third period element of the same group for example 'C*l*'. Why ? **148 P**

16. Imagine what happens lanthanoids and actinoids are inserted with in the periodic table.

(OR)

Why lanthanoids and actinoids placed seperately at the bottom of the periodic table. **148 P**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

17. Consider two elements 'A' (Atomic number 17) and 'B' (Atomic number 19) :
    i) Write the position of these elements in the modern periodic table giving justification.
    ii) Write the formula of the compound formed when 'A' combines with 'B'.
    iii) Bond formed between the two elements.
    iv) Write the electronic configurations of elements 'A' and 'B'.

18. Observe the chemical elements in the given table and answer the questions given below the table.

H	Ne	Ag	A*l*
Na	Cl	O	C
Si	F	N	Ca
Li	Ar	Zn	Xe

    i) Write the name of the element which is low reactive metal.
    ii) Write the name of the element which is metalloid.
    iii) Write the name of the element which belongs to chalcogen family.
    iv) Write name of the element which lose + 3 electrons.

19. The position of eight elements in the Modern Periodic Table is given below where atomic numbers of elements are given in the parenthesis.

Period No.	2	3	4	5
Element	Li (3)	Na (11)	K (19)	Rb (37)
	Be (4)	Mg (12)	Ca (20)	Sr (38)

i) Write the electronic configuration of Ca.

ii) Predict the number of valence electrons in Rb.

iii) What is the number of shells in Sr ?

iv) Predict whether K is a metal or a non - metal.

20. Observe the following table.

Element	A	B	C	D	E
Distribution of electrons in different shells	(2,8)	(2, 6)	(2, 8, 1)	(2 , 7)	(2, 8, 8, 2)

Now answer the following questions.

i) Write the electronic configuration and name of the element 'C'.

ii) Which element belongs to 4^{th} period ? How can you say ?

iii) Which element is inert gas ?

iv) Name the metal oxides that can be formed from the above elements.

21.

Period	Element (Atomic radius in pm)
2^{nd} period	Li(152), Be(111), B(88), C(77), N(74), O(66), F(64)
3^{rd} period	Na(186), Mg(160), A*l*(143), Si(47), P(40), S(104), C*l*(99)

Observe the above table and answer the questions.

i) What is the trend of ionisation energy from Li to F?

ii) Which elements are metals in the given table?

iii) What is the outermost orbit of 3^{rd} period elements?

iv) Among Na and Mg whose atomic radii is maximum. Why?

22. Observe the table and answer the following questions

Group No.(Name)	Electron affinity in Kj mol^{-1}
VIIA (Halogens)	F(-328); C(-349); Br(-325); I(-295); At(-270)
VIA (Chalcogens)	O(-141); S(-200); Ge(-195); Te(190); Po(-174)

i) Explain how electron affinity changes in a group from top to bottom?

ii) Explain how Electron Affinity changes in period from left to right?

iii) What does the -ve sign in the table indicates?

iv) Write any 2-Factors which influences the Electron affinity

IV. APPRECIATION AND AESTHETIC SENSE AND VALUES (AS_6) :

23. How do you appreciate the role of electronic configuration of the atoms of elements in periodic classifications ? **144, 145 P**

(OR)

How can appreciate the role of modern periodic table in understanding the properties of the elements.

V. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

24. The densities of calcium (Ca) and barium (Ba) are 1.55 and 3.51 gcm^{-3} respectively. Based on Dobereiner's law of triads can you give the approximate density of strontium (Sr) ? **137 P**

25. Why Mendeleeff has to leave certain blank spaces in his periodic table. What is your explanation for this. **139 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Mendeleef could arrange the elements similar to the arrangement of elements in modern periodic table without knowning the electron configuration of atomic elements. How do you explain this?

2. Explain the saliant features and achivements of the Mendaleef's periodic table?

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

3. Which one between Na and Cs would have less ionization energy. Guess the reason ?

4. Guess what happens if periodic table was not discovered?

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

5. Observe the table and answer the following.

Element	Atomic size (pm)	Ionizaiton energy (kJ/mol)
Li	152	520.2
Mg	160	737.7
O	66	1314
Al	143	577.5

i) In which element the removing of electron is easy ? 'Li' or 'Mg' ?

ii) Among the elements in table which has largest atomic size.

iii) 'O' has lowest atomic radius and highest ionization energy. Give the reason.

iv) Classify above elements into metals and non metals.

6. The following table indicates the atomic radii of elements belongs to second period of the periodic table. Observe the table and answer the following questions ?

2^{nd} period elements	B	Be	O	N	Li	C
Atomic radii	88	111	66	74	152	77

i) Write the decreasing order of atomic radii of above elements ?
ii) Among the 2^{nd} period element whose electronic configuration is very close to inert gas electronic configuration ?
iii) What is the outer most orbital of the above all elements ?
iv) Among the Beryllium and Carbon whose atomic radii is maximum ? Why ?

7.

Group → Period ↓	1	2	13	14	15	16	17	18
3	X		B	C	D	E		
4	Y							
5	Z							

By using above periodic table answer the following questions.
i) Name the element having smallest atomic size.
ii) Write the electronic configuration of B and E elements.
iii) Identify the elements whose physical and chemical properties are similar to element 'Y'.

8. Read the given table and answer the given questions

Group Number	Element family	Valence shell configuration	Valence electrons	Valency
IA	Alkali metal family	ms^1	1	1
IIA	Akali earth metal family	ns^2	2	2
IIIA	Boron family	ns^2np^1	3	3
IVA	Carbon family	ns^2np^2	4	4
VA	Nitrogen family	ns^2np^3	5	3
VIA	Oxygen family or Chalcogen family	ns^2np^4	6	2
VIIA	Halogen family	ns^2np^5	7	1
VIIIA	Noble gas family	ns^2np^6	8	0

i) Write the name of the element family, having "ns^2np^5" as its valance shell configuration?
ii) What is the valancy of Noble gas family elements?
iii) Write the valance shell configuration of Boron family?
iv) Why the oxygen family elements are called as chalcogen family?

9.

Group / Periods	Name of the elements (Atomic No.)							
	IA	II A	III A	IVA	VA	VI A	VIIA	VIII A
II	Li (3)	Be (4)	B (5)	C (6)	N (7)	O (8)	F (9)	Ne (10)
III	Na (11)	Mg (12)	Al (13)	Si (14)	P (15)	S (16)	Cl (17)	Ar (18)
IV	K (19)	Ca (20)	Ga (21)	Ge (32)	As (33)	Se (34)	Br (35)	Kr (36)

Answer the following from the above information.

i) Which element posses the higher atomic radius in the above table ?

ii) Mention two pair of elements which forms ionic bond.

iii) Name the two elements having valency 2.

iv) Which element has electronic configuration of $1s^2\ 2s^2\ 2p^4$.

10.

Periodic property	**Trend in**	
	Groups From top to bottom	**Periods From left to right**
Atomic radius	Increasing	Decreasing
Metallic nature	Increasing	Decreasing
Ionisation energy	Decreasing	Increasing
Electronegativity	Decreasing	Increasing
Oxidising property	Decreasing	Increasing

i) Among IA and IIA which group elements show heighest metallic nature

ii) Among IA and VIIA what is the tendency of electronegativity nature.

iii) Metallic nature increases from top to bottom of a group then what is the tendency of electropositivity in a group.

iv) In periodic table which group elements shows heighest metallic nature.

IV. APPRECIATION AND AESTHETIC SENSE, VALUES (AS_6) :

11. Appreciate different scientist in developing modern periodic table?

12. Without knowing the electronic configurations of the atoms of elements Mendeleev still could arrange the elements nearly close to the arrangements in the Modern periodic table. How can you appreciate this? **140 P**

(OR)

We can appreciate the "Mendeleef classification of elements" will you agree? Explain with correct reasons

V. APPLICATION TO DAILY LIFE, CONCERN TO BIODIVERSITY (AS_7) :

13. What are the advantages of classification of elements into s, p, d, f - blocks.

5. CHEMICAL BONDING

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS OF CONCEPTS (ROC)

1. Explain the difference between the valence electrons and the covalency of an element. **9 P (AS$_1$)**

2. A chemical compound has the following Lewis notation : **9 P (AS$_1$)**

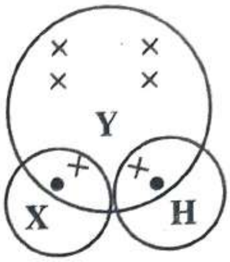
   a) How many valence electrons does element Y have ?

   b) What is the valency of element Y ?

   c) What is the valency of element X ?

   d) How many covalent bonds are there in the molecule ?

   e) Suggest a name for the element X and Y. **(AS$_2$)**



3. How bond energies and bond lengths of molecules helps us in predicting their chemical properties ? Explain with examples. **9, 10 P (AS$_1$)**

4. Draw simple diagrams to show the arrangement of valance electrons in the following compounds. **11 P**

   a) Calcium oxide (CaO)

   b) Water (H_2O) molecule

   c) Cl_2

5. Represent each of the following atoms using Lewis notation : **11 P**

   a) Beryllium b) Calcium c) Lithium d) Bromine

   e) Calcium chloride f) Carbon dioxide

6. What is hybridisation ? Explain the formation of the following molecules using hybridisation. **34 P (AS$_1$)**

   a) $BeCl_2$ b) BF_3

APPLICATIONS OF CONCEPTS (AOC)

1. Explain the formation of sodium chloride and calcium oxide on the basis of the concept of electron transfer from one atom to another atom. **(OR)** **12 P (AS$_1$)**

   How molecules will get stability based on transfer of electrons. Explain with example.

2. A, B and C are three elements with atomic number 6, 11 and 17 respectively. **13 P (AS$_1$)**

   i) Which of these cannot form ionic bond ? Why ?

   ii) Which of these cannot form covalent bond ? Why ?

   iii) Which of these can form ionic as well as covalent bonds ?

3. How Lewis dot structure helps in understanding bond formation between atoms ? **14 P(AS$_6$)**

4. Explain the formation of the following molecules using valence bond theory. **(AS$_1$,AS$_5$)**

   a) N_2 molecule - **14 P, April 23** b) O_2 molecule - **15 P**

HIGHER ORDER THINKING QUESTIONS (HOTS)

1. Two chemical reactions are described below. 15, 16 P (AS_5)

   Nitrogen and hydrogen react to form ammonia (NH_3)

   Carbon and hydrogen react to form a molecule of methane (CH_4)

   For each reaction, give

   a) The valency of each of the atoms involved in the reaction

   b) The Lewis structure of the product that is formed.

SUGGEST PROJECTS :

1. Collect the information about properties and uses of covalent compounds and prepare a report.

   16, 17 P (AS_4)

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Write the 'Octet Rule'. How does Mg (12) get stability while reacting with chlorine as per this rule? **25, 26 P(2015 Public)**
2. Explain the formation of BF_3 molecule with the help of VSEPR Bond Theory. **12 P (2018 Public)**
3. Explain the formation of N_2 molecule by using valence bond theory. **14 P (2023 Public)**
4. Define polar covalent bond and explain formation of polar covalent bond in HC*l* molecule. **32 P (2023 Public)**

II. INFORMATION SKILLS AND PROJECTS (AS_4) :

5.

Element	Z	Electronic configuration				Valence
		K	L	M	N	
Helium (He)	2	2				2
Neon(Ne)	10	2	8			8
Argon(Ar)	18	2	8	8		8
Krypton(Kr)	36	2	8	18	8	8

Based on above information answer the questions : **(2017 Public)**

i) Innert gases are highly stable and unreactive justify this statement.

ii) Helium contains only two electrons but it is treated as innert gas. Justify this statement.

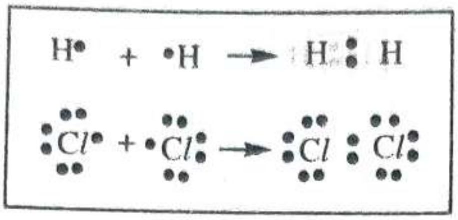
iii) Why innert gases are kept at the end of the periodic table. Explain its advantage.

iv) What is the maximum number of electrons accomdated in each shell.

III. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

6. Draw the structural diagram of Ammonia molecule as per the valence-shell electron pair repulsion theory. **36 P (2016 Public)**

7. Show the formation of HCl molecule with Lewis dot structures using the information given below.



(2017 Public)

8. Draw the structure of Methane molecule and mention bond angle. **33 P (2019 Public)**

IV. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

9. Write any two uses and two properties of Covalent compounds. **16, 17 P (2015 Public)**

10. A, B are two elements. Its compound material is A_2B, then what will be the valency of A and B. **(2018 Public)**

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. What is hybridisation ? Explain the formation of the following molecules using hybridisation.
   a) NH_3 - **36 P** b) H_2O - **37 P**
2. Explain electronic theory of valence by Lewis and Kossel? Write the Draw backs of electronic theory of valence. **20, 22, 32 P**
3. How VSEPR theory predicts shapes of the molecules? Write the Limitations of VSEPR theory ?
4. Why do some elements and compounds react vigorously while others are inert ? **18 P**
5. Explain how a covalent bond is formed with an example? **21 P**
6. List the factors that determine the type of bond that will be formed between two atoms ? **24 P**
7. Why do only valence electrons involve in the bond formation ? Why not electron of inner shells ? Explain. **21 P**
8. Sodium atom lose one electron and forms cation, chlorine atom gain one electron and forms anion. What are the factors that influence the formation of cation and anion. **12 P**
9. With four bond pairs $H\hat{C}H$ bond angle in CH_4 is $109°28'$, with three bond pairs and one lone pair $H\hat{N}H$ bond angle in NH_3 is $107°48'$, with two bond pairs and two lone pairs $H\hat{O}H$ bond angle in H_2O is $104°31'$. Conclude all these statements with suitable explanation and diagrams. **34, 35, 36 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

10. Why noble gases are chemically inert ? Guess and write? **18 P**
11. Predict the reasons for low melting point for covalent compounds when compared with ionic compound. **27 P**

12. Sigma bond is stronger than pi bond. Predict the reason. **33 P**

13. Predict and write "bond angle" of $BeCl_2$ is 180°, why in H_2O is 104°31?

14. Why is the chemical formula for water H_2O and for sodium chloride NaC*l*, why not HO_2 and $NaCl_2$. **22 P**

15. 'Like dissolve in like' Justify this statement with suitable examples. **25 P**

16. Guess the reasons why Ionic compounds exhibit high Melting Point or Boiling point

(OR)

Explain why ionic compounds exhibit more M.p and B.p than covalent compounds.

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

17. The electronic configuration of X and Y atoms are $1s^2\ 2s^2\ 2p^6\ 3s^1$ and $1s^2\ 2s^2\ 2p^4$ then answer the following questions.
    - i) Which atoms can form anion? Why ?
    - ii) Which atoms can form cation? Why?
    - iii) Name the blocks of X,Y elements in the periodic table ?
    - iv) Write the molecular formula which is formed by atoms of elements X and Y ?

18.

Element	$_1H$	$_{11}Na$	$_{17}Cl$	$_{18}Ar$
Electrons in atomic orbital	1	2, 8, 1	2, 8, 7	2, 8, 8

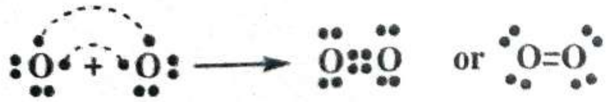
Based on the above table, answer the following questions.
- i) Which element has '8' electrons in the outer most orbit?
- ii) Represent the formation of NaC*l* molecule using lewi's dot structure.
- iii) If Sodium loses one electron, it attains which inert gas configuration?
- iv) Represent the valence electron of Hydrogen using lewi's dot structure.

19. Analyze the following table and answer the below given questions

Element	A	D	E
Valency	1	2	1
No. of valence electrons	7	6	1

- i) Draw Lewis electron dot structures of elements A, D, E.
- ii) Classify elements as metals and non-metals.
- iii) Write the chemical formulae of possible Ionic substances from the given table of elements.
- iv) Write the element of IA (1^{st}) group of periodical table.

20. Observe the formation of oxygen molecule and answer the following questions.



Questions :

i) How many bonds between oxygen atoms?

ii) Mention the valency of oxygen atom.

iii) What type of bond does oxygen molecule has?

iv) How many valence electrons are there in oxygen atom?

21. a) $\ddot{\mathrm{He}}$ b) $:\underset{\cdot\cdot}{\ddot{\mathrm{Ne}}}:$ c) $:\underset{\cdot\cdot}{\ddot{\mathrm{Ar}}}:$ d) $:\underset{\cdot\cdot}{\ddot{\mathrm{Kr}}}:$ e) $:\underset{\cdot\cdot}{\ddot{\mathrm{Xe}}}:$ f) $:\underset{\cdot\cdot}{\ddot{\mathrm{Rn}}}:$

By observing the above structrues answer the following questions. **153 P**

i) What is the name of the above structures

ii) Name the rule involved in drawing above structures

iii) What is the valency of element Helium.

iv) Among the given elements which can form compounds at perticular conditions ?

22.

S.No.	Molecule	No.of bond pair of electrons	No.of lone pairs of electrons
1	BF_3	3	0
2	NH_3	3	1
3	H_2O	2	2

Observe the above table and answer the following questions.

i) What is the bond angle in BF_3 molecule ?

ii) What is the shape of water molecule ?

iii) What is the type of hybridisation of 'B' in the NH_3 molecule?

iv) How many number of lone pairs are present on oxygen in H_2O molecule?

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

23. Draw the structures of water molecule H_2O 'V-shape' and Ammonia molecule NH_3 trigonal pyramid. **37 P**

24. Show with a diagram, the bond present in an Oxygen molecule. What type of bond is present in that? **15 P**

(OR)

Draw a neat diagram of formation of Oxygen (O_2) or Nitrogen(N_2) molecule. (using V.B.T)

25. Draw and communicate the formation of following molecules and show how electrons are arranged in them based on Lewis notation.

a) NaCl **12 P** b) MgCl **25 P** c) CH_4 **28 P** d) NH_3 **29 P**

V. APPRECIATION AND AESTHETIC SENSE AND VALUES (AS_6) :

26. What is octet rule ? How do you appreciate role of the 'octet rule' in explaining the chemical properties of elements ? **12 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Explain the formation of the following molecules using valence bond theory.
   a) H_2 molecule - **35 P** b) Cl_2 molecule - **33 P**
2. Differentiate properties of ionic, covalent and polar covalent compounds **25 P**
3. Explain the formation of molecules with four bond pairs based on VSEPR theory. **33 P**
4. Why do some elements exist as molecules and some as atoms ? **18 P**
5. How do cations and anions of an ionic compound exist in its solid state ? **24 P**
6. Explain the formation of following molecules based on VSEPR theory.
   i) molecule with two bond pairs and no lone pairs **29 P**
   ii) molecule with three bond pairs and no lone pairs **29 P**
   iii) molecule with three bond pairs and one lone pair **36 P**
   iv) molecule with two bond pairs and two lone pairs **39 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

7. Imagine and write what happend if VSEPR thory was not proposed and developped by scienctists?

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

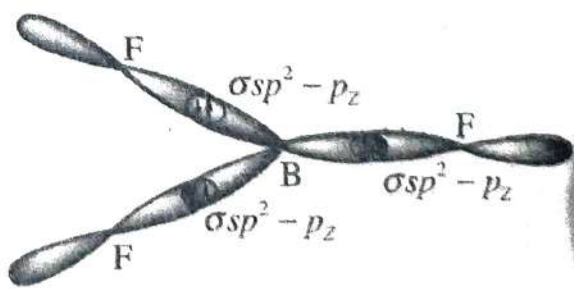
8. Observe the following table and answer the questions given below.

Bond	H – H	I – I	H – O	Br – Br	H – Cl
Bond length (A°)	0.74	2.68	0.96	2.28	1.27

   i) What is Bond length H – O in water moleule ?
   ii) Why I – I bond length is more than H – H bond length ?
   iii) What are units of bond length ? How many nanometers are equal to 1A° ?
   iv) If bond length increases, what happens to bond energy.

9. An element 'X' donates 2 electrons to the element Y and forms the compound 'XY'. The element X with the element Z froms a compound XZ_2.
   i) Write the valencies of the elements X, Y and Z.
   ii) Name the type of bond formed between X and Y
   iii) The solvent in which the compound 'XY' is soluble ?
   iv) What are the valencies of X, Y and Z?

10. Answer the following by observing the information.



   i) Name the molecule and write formula.
   ii) What is the bond between B and F ?
   iii) Name the hybridized orbitals in Boron.
   iv) What is the bond angle in the above molecule according to VESPR theory ?

11.

S.No.	Property	NaC*l*(ionic)	HC*l*(polar covalent)	C_2H_6(Covalent)
1.	Formula mass	58.5	36.5	30.0
2.	Type of bond	Ionic	Polar covalent	Covalent
3.	Melting point	801°C	–115°C	–183°C
4.	Boiling point	1413°C	–84.9°C	–88.63°C
5.	Solubility	______	Soluble in polar solvents like water and to some extent in non-polar solvents	______
6.	Chemical activity	______	Moderately reactive	Slow or very slow at room temperature

25 P

By analysing the table fill the gaps and answer the following questions.

i) Why ionic compounds are soluble in water.

ii) Why chemical reactivity of covalent compounds is slow.

iii) How can you say HC*l* contains polar covalent bond.

iv) Arrange NaC*l*, HC*l* and C_2H_6 in increasing order of its melting point.

IV. COMMUNICATION THROUGH DRAWING, MODEL MAKING (AS_5)

12. Draw the neat diagram of Sodium chloride.(FCC structure)

(OR)

Draw and communicate on the Lattice structure of NaCl molecule. **23 P**

(OR)

Show the arrangement of ions in ionic compounds.

13. Represent the following molecule and their bond formation based on Lewis notation.

a) Br_2 **30 P** b) CO_2 **30 P** c) Na_2O **26 P**

Identify which of the above molecules will contains a double bond.

14. Represent each of the following atoms using Lewis notation :

a) Al_2O_3 **26 P** b) $CaCl_2$ **30 P** c) $AlCl_3$ - **27 P**

6. PRINCIPLES OF METALLURGY

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

1. List three metals that are found in nature as oxide ores ? **57 P (AS_1)**
2. List three metals that are found in nature in uncombined form. **57 P (AS_1)**
3. Write a note on ore dressing or concentration of ore in metallurgy ? **57 P (AS_1)**
4. How do metals occur in nature ? Give examples to any two types of minerals. **58 P (AS_1)**
5. When do we use magnetic separation method for concentration of an ore? Explain with an example. **58, 59 P (AS_1)**
6. What is the difference between roasting and calcination? Give one example for each ? **59 P (AS_1)**
7. Draw the diagram showing i) Froth floatation ii) Magnetic separation **60 P (AS_5)**
8. Draw a neat diagram of Reverberatory furnace and label it ? **(TG - 2015 public) 61 P (AS_5)**

APPLICATIONS OF CONCEPTS (AOC)

1. Magnesium is an active metal, if it occurs as a chloride in nature, which method of reduction is suitable for its extraction ? **61 P (AS_1)**
2. Mention and explain any two methods which produce very pure metals from impure metals ?(AS_1) **(TG - 2015 Sup) 61 P**
3. Which method do you suggest for extraction of high reactivity metals? Why? **62 P (AS_1)**

   (OR)

   Guess the reasons why highly rective metals like K, Na, Ca, Mg are extracted by electrolysis of their fused salts only. **2017 Public**

   (OR)

   Explain how metals presented the top of the activity series are extracted **65 P (AS_1)**

   (OR)

   How Na*Cl* is extracted by Electrolytic reduction.
4. Explain thermite process ? Mention its applications in daily life. **62, 63 P (AS_1) (2015 public)**

   (OR)

   Explain the related chemical reactions of the process where aluminium is used as a good reducing agent and the pure metal obtained is used for welding and joining railway tracks or cracked machine parts. (AS_7)

   (OR)

   $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe + Heat$ (AS_7)

   It is thermite reaction identify the situations in daily life where this reaction is used ?

5. Where do we use handpicking and washing methods in our daily life ? Give examples. How do you correlate these examples with enrichment of ore ? **63 P (AS_7)**

SUGGESTED EXPERIMENTS

1. What is corrosion Suggest an experiment to prove that the presence of air and water are essential for Corrosion. Explain the procedure. **64 P (AS_3) (TG - 2019)**

(OR)

What happens if iron articles are exposed to moist air ? Write the chemical equation to represent that reaction. **(AS_3) (TG - 2015 Supply)**

SUGGESTED PROJECT

1. Collect information about extraction of metals of low reactivity silver, platinum and gold and prepare a report. **65 P (AS_4)**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Give an example with the chemical equation for the reduction of ores using moderately reactive metals. **57, 58, 74 P (2017 Public)**

2. Which method used for concentration of sulphide ore ? Explain the process ? **(2017 Supply)**

(OR)

Mention the four physical methods of concentration of the ore and explain any two of them. **73 P (2023 Public)**

3. What are the preventive methods do you take for rusting of iron materials ? **80 P (2018 Public)**

4. Write the products of given reactions, if any. Give reasons. **(2016 Public)**

i) $FeCl_2 + Zn \rightarrow$ ii) $ZnCl_2 + Fe \rightarrow$

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

5. Predict, what happens in the field of domestic use of metals if alloy were not discovered. **80 P (2016 Supply)**

6. What would happen, if the metals like Copper and Iron do not get oxidized ? Guess and write? **(2022 Public)**

7. What happens if rusting of metals does not prevent? **80 P (2023 Public)**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

8. Observe the following table ans answer the questions **(2023 Supply)**

Name of the Ore	Bauxite	Zinc blende	Epsom salt	Cinnabar	Rock salt	Pyrolusite	Galena
Formula	$Al_2O_3.2H_2O$	ZnS	$MgSO_4.7H_2O$	HgS	NaCl	MnO_2	PbS

i) Which ores are oxides among the table?
ii) How many Sulphate ores are there in the given table? What are they?
iii) Which ore are concentrated by froth flotation method in the given table?
iv) Which ores are extracted by electrolysis method in the given table?

IV. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

9. We use P.V.C pipes for water supply instead of metal pipes. Why ? **77 P (2017 Public)**

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. "All Ores are minerals but all minerals need not to be ores". Justify the statement? **68 P**
2. Define the following terms : i) Gangue ii) Slag iii) Flux **67, 78 P**
3. How can you differentiable between minerals and ores? **68 P**
4. For metals and ores write the examples for the following. In a tabular form
i) Sulphate ii) Carbonate iii) Sulphide iv) Halide
5. What is galvanisation ? How does it protect iron from rusting ? **79 P**
6. What is rust ? Give the equation for the formation of rust ? **78 P**
7. $Fe_2O_3 + 3CO \xrightarrow{\text{in blast furnace}} 2Fe + 3CO_2$
For this reaction identify reagents undergoing / agents
i) Oxidation ii) Reduction iii) Oxidising agent iv) Reducing agent
8. What is activity series ? How it helps in extraction of metals ? **70 P**
(OR)
Give high reactive metals, moderate reactive metals and low reactive metals in their descending order?
9. Name two metals which will displace hydrogen from dilute acids, and two metals which will not.

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

10. Guess and write why Gold, Silver and Platinum are used for Jewelleries ?**(OR)**
Why do gold ornaments look new even after several years of use? **70 P**
11. Iron ore is concentrated by magnetic separation method. Why? Guess and write?

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

12. By observe the concept of corrosion answer the following questions.
i) Imagine what happens when iron articles kept in the open atmosphere ?
ii) What is the formula of "rust" ?
iii) Suggest some prevention methods for corrosion process ?
iv) Compare the advantage of alloys over the metals ?

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

13. Draw the following diagrams with neat labelling
i) Electrolytic refining of copper **76 P** ii) Electrolytic reduction of fused NaCl **62 P**

V. APPRECIATION AND AESTHETIC SENSE, VALUES (AS_6)

14. "Appreciate use of stainless steel and 22 carat gold are better to use in our daily life than Iron and 24 carat gold". **81 P**

VI. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

15. Where do we use handpicking and washing methods in our daily life ? Give examples. How do you correlate these examples with enrichment of ore? **63 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Define furnace and name the parts and their importance in a furnace. **82 P**
2. Explain corrosion mechanism with sutable exmples. **81 P**
3. Iron article are kept moist atmosphere change their colour and composition. Identify different types of substances which change their colour and composition and kept exposed to moist atmosphere.

(OR)

Vasu observed colour change on iron articles and copper articles when kept in moist atmosphere. What could be the reason for all these. Explain same type of situations in daily life. **78 P**

4. What is corrosion and what are the preventive methods corrosion
i) Stain less steel ii) Steel iii) Bronze iv) Brass

(OR)

Name three alloy steels ? Give their compositions and uses ? **79 P**

5. Why Sulphide ores need to roasting and then reduced with carbon.
6. Why highly reactive metals cannot be extracted by reduction by carbon or carbon monoxide. **76 P**
7. Out of C and CO which is a better reducing agent for FeO and where they are used properly at different parts of the blast furnace. **71 P**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS(AS_2) :

8. Predict and write any one consequences, if impurities are not added to the ore while extracting of high reactive metals.
9. Predict and write why is a reverberatory furnace better than other furnaces.
10. What happens imagine and write, if didn't concentration of ore?
11. Guess and write what happens if there is no oxidation in nature.
12. Predict and write any one consequences, if impurities are not added to the ore while extracting of high reactive metals.

II. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

13. How do various metals in activity series react with steam ? **56 P**

14. Explain an acitivity of electrolytic refining of copper(Blister copper). **76 P**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

15. **68, 69 P**

Oxides	Sulphides	Chlorides	Carbonates	Sulphates
—	—	—	—	—
—	—	—	—	—

Fill the gaps in the table with names of the respective ores and answer the following questions.

i) What do you notice in the table ?

ii) Name the methods how carbonate ores are extracted and sulphide ores are concentrated.

16. Four metals A, B, C, and D are in turn added to the following solutions one by one.

The obserations made are tabulated below. **69 P (AS_4) (TG - 2015 public)**

Metal	Iron(II) sulphate	Copper(II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement	-	-
B	Displacement	-	No reaction	-
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

Answer the following questions based on the above information :

i) Which is the most reactive metal ? Why ?

ii) What would be observed, if 'B' is added to a solution of Copper (II) sulphate and why ?

iii) Arrange the metals A, B, C, and D in order of increasing reactivity.

iv) Which one among A, B, C, and D metals can be used to make containers that can be used to store any of the above solutions safely ?

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

17. Draw the neat labelled diagram of Blast furnace. **174 P**

V. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

18. Write the daily life applications of electrolytic refining ?

19. What are the applications of electrolysis in our daily life?

7. CARBON AND ITS COMPOUNDS

IMPROVE YOUR LEARNING

PRIORITY - I

REFLECTIONS ON CONCEPTS (ROC)

1. What are the general molecular formulae of alkanes, alkenes and alkynes. **104 P (AS_1)**
2. Name the product other than water formed on burning of ethanol in air. **104 P (AS_1)**
3. Name the simplest ketone and write its molecular formula. **105 P (AS_1)**
4. Name the compound formed by heating ethanol at 443 K(170°C) with excess of conc. H_2SO_4. **105 P (AS_1) (T & D)**
5. Name the product obtained when ethanol is oxidized by either chromic anhydride or alkaline potassium permanganate. **105 P (AS_1)**
6. Write the homologous series of carbon compounds after $CH_3OHCH_2CH_3$ and write the IUPAC name. **106 P**
7. Why does Carbon form compounds mainly by covalent bonding ? **106 P (AS_1)**
8. Explain how sodium ethoxide is obtained from ethanol ? Give chemical equations. **106 P (AS_1)**

   (OR)

   What happens when a small piece of sodium is dropped into ethanol ?
9. Explain the cleansing action of soap. **107 P (AS_1)**

   (OR)

   Explain the mechanism of cleansing action of soaps. Define saponification. **108 P**
10. Distinguish between esterification and saponification reactions of organic compounds. **108 P (AS_1)**
11. Draw the electronic dot structure of ethane molecule (C_2H_6) **108 P (AS_5)**

APPLICATIONS OF CONCEPTS (AOC)

1. Explain with the help of a chemical equation, how an addition reaction is used in vegetable ghee industry. **109 P (AS_1)**
2. i) What are the various possible structural formulae of a compound having molecular formula C_3H_6O **109 P (AS_1)**

   ii) Give the IUPAC names of the above possible compounds and write their structures. **(AS_1)**

   iii) What is the similarity in these compounds ? **(AS_1)**
3. Allotropy is a property shown by which class substance : Elements, Compounds or mixtures ? Explain allotropy with suitable examples. **110 P (AS_1)**
4. Two carbon compounds A and B have molecular formula C_3H_8 and C_3H_6 respectively. Which one of the two is most likely to show addition reactions ? Justify your answer. **110 P (AS_2)**
5. 1 ml of glacial acetic acid and 1 ml of ethanol are mixed together in a test tube. Few drops of concentrate sulphuric acid is added to the mixture are warmed in a water bath for 5 min. **111 P (AS_1)**

Answer the following :

a) Name the resultant compound formed.

b) Represent the above change by a chemical equation.

c) What term is given to such a reactions.

d) What are the special characteristics of the compound formed ?

SUGGESTED EXPERIMENTS

1. Suggest a test to find the hardness of water and explain the procedure. **111, 112 P (AS_3)**
2. Suggest a chemical test to distinguish between ethanol and ethanoic acid explain the procedure.

   A students is unable to detect solutions of ethyl alcohol and acetic acid which are present side by side bottles in a lab identify these two chemicals with Na_2CO_3 or $NaHCO_3$. **113 P (AS_3)**

   (OR)

   How can ethanol and ethanoic acid can be differentiated on the basis of their physical and chemical properties? (AS_3)
3. An organic compound 'X' with a molecular formula C_2H_6O undergoes oxidation with alkaline $KMnO_4$ and forms the compound 'Y', that has molecular formula $C_2H_4O_2$. **113 P (AS_3)**

   a) Identify 'X' and 'Y'

   b) Write your observation regarding the product when the compound 'X' is made to react with compound 'Y' which is used as a preservative for pickles.

SUGGESTED PROJECTS

1. Collect information about artificial ripening of fruits by ethylene. **114 P (AS_4)**
2. How do you condemn the use of alcohol as a social practice. **114 P (AS_7)**

PREVIOUS QUESTION PAPERS

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. How do you explain the role of Oxygen in combustion process ? **(2015 Public)**
2. Write the types of Allotropes of Carbon. Give any three examples of each. **(2016 Public)**
3. What are homologous series of carbon compounds ? Mention any four characteristics of homologous series. **123 P (2016 Public)**
4. What are alkenes ? Write the general formula of alkenes. Give an example for alkenes. **(2017 Supply)**
5. Explain the isomerism and Catenation properties of carbon. **(2018 Public)**
6. Why do we call alkanes as paraffins ? Explain the substitution reactions of alkanes. **140 P (2022 Public)**
7. Mention the differences between Alkanes and Alkenes. **123 P (2023 Public Model)**

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

8. $CH_2 = CH_2 + H_2 \xrightarrow{Ni} CH_3 - CH_3$ is an addition reaction. $CH \equiv CH + H_2 \xrightarrow{Ni}$?

   Predict and write the products. **(2016 Public)**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

9. List out the materials required to conduct the experiment to understand the esterification reaction. Explain the procedure of the experiment. How can you identify that an ester is formed in this reaction ? **143 P (2017 Public)**

(OR)

Write the required materials, chemicals for "esterification reaction" activity. **143 P (2023 Public)**

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

10. In the table given below, fill the information in the empty boxes and give answer to the following questions. **(2015 Supply)**

S.No.	Alkane	Molecular formula	Structure	Number of Carbons
1)	Methane	CH_4	-	1
2)	Ethane	C_2H_6	$-\overset{\vert}{\underset{\vert}{C}}-\overset{\vert}{\underset{\vert}{C}}-$	2
3)	Propane	C_3H_8	$H-(CH_2)_3-H$	3
4)	Butane	C_4H_{10}	$H_3C-CH_2-CH_2-CH_3$	4

i) Write the general formula of alkanes from the table.
ii) How many σ bonds are there in C_2H_6.
iii) What sequential order did you notice in the molecular formulae?
iv) There exists single bonds between carbon atoms of alkanes. do you agree with this statement? Give reasons.

11.

Organic compound	Methane	Ethane	Propene	Butene	Pentyne	Hexyne
Formula	CH_4	C_2H_6	C_3H_6	C_4H_8	C_5H_8	C_6H_{10}

Observe the above table and answer the following questions. **(2019 Public)**
i) Write the general formula of Alkanes.
ii) Mention the names of unsaturated hydrocarbons.
iii) Write the homologous series of Alkynes.
iv) Write the formula of Hexyne.

V. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

12. Draw the structure of the following carbon compound. **(2019 Public)**
3, 7-dibromo-4, 6-dichloro - oct-5-ene-1, 2-diol.

VI. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

13. Write two uses of Ethanol in day to day life. **141 P (2018 Public)**

14. Mention uses of Nanotubes which is an allotrophy form of Carbon in day to day life. **117 P (2017, 2023 Public)**

PRIORITY - II

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Explain the structure of diamond and Graphite. 117 P
2. "All combustion reactions are not oxidation reactions " Justify the statement. 143 P
3. Define soap, micelle and "cmc" ? 144 P
4. What are structural isomers? Draw structural isomers of Butane (C_4H_{10}). 124 P
5. Describe with chemical equation how ethanoic acid may be obtained from ethanol.

   (OR)

   "Oxidation of ethanol $\rightarrow$ ethanal $\rightarrow$ ethanoic acid" explain ? 136 P
6. Define the following terms ? Give one example for each. 116 P
   i) Hybridisation ii) Catenation iii) Oxidation iv) Functional group
7. Explain the following molecules by hybridization :
   i) Methane (sp^3) - **119** ii) Ethane(sp^3) - **(TB)** iii) Ethyne (sp) - **121** iv) Ethene (sp^2) - **120**
8. Write the chemical equation representing the reaction of preparation of ethanol from ethene.

   (OR)

   How will you convert ethene into ethanol ? Give the chemical reaction involved ? **111 P**
9. Define saturated and unsaturated hydrocarbons? **124 P**

   (OR)

   Compare the physical and chemical properties of alkanes, alkenes and alkynes.
10. List any four differences between soaps and detergents. **(OR)** **135 P**

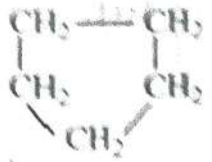
    What is the difference between the chemical composition of soaps and detergents ? **141 P**
11. Write the physical and chemical properties of acetic acid or ethanoic acid ? **142 P**
12. Name the functional group present in each of the following organic compounds
    i) C_2H_5Cl ii) C_2H_5OH iii) HCOOH iv) C_2H_5CHO
    v) CH_3COCH_3 vi) C_2H_5COOH **128, 129 P**
13. How would you name the following compounds ?

    i) CH_3-CH_2-Br ii) $CH_3-CH_2-CH_2-CH_2-C\equiv CH$

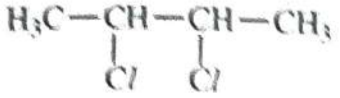
    iii) $CH_3-\overset{\displaystyle H}{\overset{|}{C}}=O$ iv) $CH_3-CH_2-CH_2-COOH$

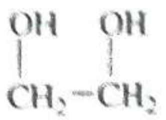
14. Give the IUPAC name of the following compounds.

a) $CH_3 - CH_2 - CH = CH - CH_3$

b) 

c) $CH_3 - CH_2 - CH_2 - CH_2 - CHO$

d) 

e) 

f) $HC \equiv C - CH = CH_2$

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

15. Why do some times cooking vessels get blackened on a gas or kerosene stove? **132 P**

16. "Saturated hydrocarbons burn with a blue flame while unsaturated and aromatic hydrocarbons burn with a sooty flame." Why? **(OR)** **131 P**

    Why aromatic compounds will produce sootyflame ? **136 P**

III. EXPERIMENTATION AND FIELD INVESTIGATION (AS_3) :

17. Give a test to identify the presence of ethanoic acid ? **139 P**

IV. INFORMATION SKILLS AND PROJECTS (AS_4) :

18. An organic compound with molecular formula $C_2H_4O_2$ produces brisk effervescence on addition of sodium carbonate / bicarbonate. Answer the following : **132 P**
    - i) Identify the organic compound
    - ii) Write the chemical equation for the above reaction.
    - iii) Name the gas evolved.
    - iv) How will you test the gas evolved ?

19. Observe the given below table and answer the questions below the table.

S.No.	General formula of hydro carbons	Name of the hydro carbons
1.	C_nH_{2n+2}	Alkane
2.	C_nH_{2n}	Alkene
3.	C_nH_{2n-2}	Alkyne
4.	C_nH_{2n+1}	Alkyl

Questions :

- i) Which hydro carbons have triple bond between carbon atoms?
- ii) Write the formula for Butene.
- iii) Write the structure for Pentyne.
- iv) Write the name of hydrocarbon C_6H_{13}.

20. Observe the following table.

Name of Hydrocarbon	Methane	Ethylene	Acetylene	Butene	Propyne	Ethane
Formula	CH_4	C_2H_4	C_2H_2	C_4H_8	C_3H_4	C_2H_6

Now answer the following questions.

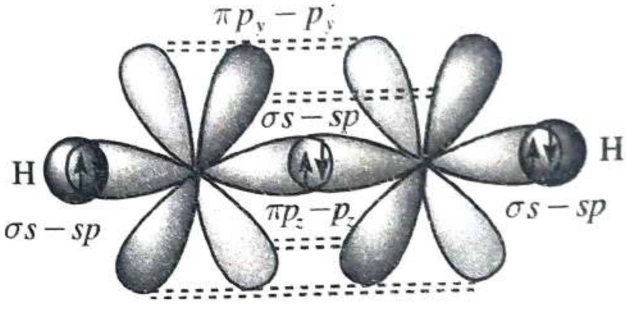
i) Identify the Primary Alkene.

ii) Name the saturated Hydrocarbon.

iii) Name the compounds having triple bond.

iv) Write any two Homologs from the table.

21. By observing the following bond formation in ethene, answer the following questions.



Ethyne or Acetylene

i) What type of hybridization is present in the molecule?

ii) What type of sigme bond is present between carbon - hydrogen atoms?

iii) How many bonds are formed between carbon - carbon atoms? What are they?

iv) Draw the structural formula of the above molecule.

V. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

22. Write the structure of the compound 4-bromo-7,8-dichloro-5-methyl oct-1,2-diene-1,3-diol.

23. Draw the following diagrams with neat labelles
i) Micelle formation ii) Soap **(OR)** **142, 144P**

Draw a diagram of soap particle showing. hydrophobic end and hydrophilic end.

24. Draw the hybridisation structures of the following compounds :
i) C_2H_2 **- 121 P** ii) C_2H_6 iii) C_2H_4 **- 120 P** iv) CH_4 **- 119 P**

VI. APPRECIATION AND AESTHETIC SENSE AND VALUES (AS_6) :

25. Appreciate the versatile nature of carbon in forming large number of carbon compounds.

26. How do you appreciate the role of esters in everday life **137 P**

VII. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

27. Write uses of ethanoic acid or acetic acid ? **139 P**

28. Write about Buckminsterfullerene ($_{60}C$) and who discovered it. **121, 122 P**

29. Applications of Graphite and Graphene.

30. Write the Allotropes of carbon? And write its applications in our daily life? **118 P**

PRIORITY - III

I. CONCEPTUAL UNDERSTANDING (AS_1) :

1. Explain fermentation (or) Explain the process of obtaining ethanol from starch. **(TB)**
2. How can you distinguish saturated and unsaturated hydrocarbons with flame test. **13P**
3. Why graphite act as good conductor of electricity and pencil marking can be easily erased ? **115 P**
4. What is a catalyst? Write the chemical equation to represent the hydrogenation of ethene? **139 P**
5. A mixture of oxygen and ethyne is burnt for welding ; can you tell why a mixture of ethyne and air is not used ? **133 P**
6. Compare how primary and secondary amines are obtained from NH_3 by replacing hydrogen as like in R – O – R, R – O – H from H_2O. **129 P**
7. How the police detect whether suspected drivers have consumed alcohol or not? **142 P**
8. Write the formula of the compound and name of the functional groups present in each of them:
   i) Ethanoic acid ii) Propanone iii) Nitromethane **128 P**
9. Complete the following equations and write the name of the products formed.

   i) $CH_3COOH + NaOH \xrightarrow{\text{heat}}$ ii) $C_2H_5OH + O_2 \xrightarrow[\text{KMnO}_4]{\text{Alkaline}}$ **131, 136 P**

   iii) $CH_3COOH + C_2H_5OH \xrightarrow{\text{Conc } H_2SO_4}$

   iv) $CH_3COOH + Na_2CO_3 \longrightarrow$
10. Complete the following reactions **137 P**

    i) $CH_3CH_2OH \xrightarrow[heat]{Conc. H_2SO_4}$ ii) $CH_3COOH + NaHCO_3 \rightarrow$

    iii) $CH_4 + Cl_2 \xrightarrow{Sunlight}$ iv) $CH_2 = CH_2 + H_2 \xrightarrow{Ni}$

II. ASKING QUESTIONS AND MAKING HYPOTHESIS (AS_2) :

11. Why only oil is recommended for cooking but not animal fat.
12. Imagine and write what happend if cloths are washed with Hard water with soap? **135 P**

III. INFORMATION SKILLS AND PROJECTS (AS_4) :

13. Represent the 11 disginations in IUPAC nomenclature.

Numbers - Numerical Prefixes - Secondary Prefix - Primary Prefix - Word Root - Numbers 1 2 3 4 5 6
- Numerical Prefix, Primary Suffix, numbers, numerical Prefixes and Secondary Suffixes 7 8 9 10 11

Questions

i) What is the information gives by secondary suffix?

ii) What is difference between Numerical prefixes with designations 7 and 2.

iii) While writing IUPAC names of compounds which designation will give information regarding saturation and unsaturation of compound.

iv) While writing IUPAC names of compounds which designation will give information regarding the principle functional group.

14. Answer the following question by using the above data

Formula	IUPAC Name
C_2H_6	Ethane
C_2H_4	Ethene
C_2H_2	Ethyne
CH_3COOH	Ethanoic acid
C_2H_5OH	Ethanol
C_2H_5CHO	Propanal

i) Write the unsaturated hydrocarbons.

ii) Identify the compounds of aldehyde functional group.

iii) What is the common name of Ethanoic acid.

iv) Write the general formula of alkanes.

IV. COMMUNICATION THROUGH DRAWING AND MODEL MAKING (AS_5) :

15. Draw the following diagrams with neat labelles
i) Diamond - **116 P** ii) Graphite - **118 P**

16. Draw the neat labelled of diagram with Buckminster fullerene(C_{60}). **122 P**

17. Draw the structures of the following compounds :
i) Ethanoic acid ii) Bromopentane iii) Butanone iv) Hexanal

18. Draw the electron dot structures for **132 P**
i) Ethanoic acid ii) Propanone iii) Butanaldehyde iv) Ethyl alcohol

V. APPLICATION TO DAILY LIFE AND CONCERN TO BIODIVERSITY (AS_7) :

19. What are the affects of Oxidation on everyday life ?

S.S.C PUBLIC EXAMINATIONS, MARCH : 2023 - 24

19E(A)

SCIENCE

Part - I : Physical Science

(English version)

Parts A and B

Time : 1 hour 30 mins] **[Max. Marks : 40**

Instructions :

i) Read the question paper and understand every question thoroughly and write answers in given 1.30 min. time.

ii) **3 very short answer questions** are there in **Section - I**. Each questions carries **2 marks**. **Answer all the questions**. Write answer to each question in **3 to 4 sentences**.

iii) **3 short answer questions** are there in **Section - II**. Each question carries **4 marks. Answer all the questions**. Write answer to each question in **5 to 6 sentences**.

iv) **3 essay type answer** quetions are there in **Section - III**. Each question carries **6 marks**. Answer **any two** questions. Write answer to each question in **8 to 10 sentences**.

PART - A

Time : 1 hour 15 minutes **Marks : 30**

SECTION - I

$3 \times 2 = 6$

Instructions :

i) **3 very short answer** questions are there in this **Section - I**.

ii) **Answer all the questions.** Each question carries **2 marks.**

iii) Write answer to each question in **3 to 4 sentences.**

1. Write any two daily life uses of Baking soda.
2. Write the required materials, chemicals for "esterification reaction" activity.
3. Guess and write, what changes may happen to the focal length and position of optic centre of Convex lence, if it is broken into two pieces?

SECTION - II

$3 \times 4 = 12$

Instructions :

i) **3 Short answer** questions are there in this **Section - II**.

ii) Answer **all** the questions. Each question carries **4 marks.**

iii) Write answer to each question in **5 to 6 sentences.**

4. Mention the four physical methods of concentration of the ore and explain any two of them.

5. Write any four applications of Faraday's "Electromagnetic induction", and explain any two of them.

6. Observe the information given in the table. Answer the questions given below :

Element	Electronic configuration
Be	$1s^2$ $2s^2$
Mg	$1s^2$ $2s^2$ $2p^6$ $3s^3$
P	$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^3$
Ne	$1s^2$ $2s^2$ $2p^6$

i) Which of the given are s - block elements?

ii) Which of the given has least valency?

iii) Which of the given is 15^{th} Group element?

iv) Which of the given are of same Period?

SECTION - III

$2 \times 6 = 12$

Instructions :

i) **3 Essay type** questions are there in this **Section**.

ii) Answer any **two** the questions. Each question carries **6 marks.**

iii) Write answer to each question in **8 to 10 sentences.**

7. LDraw the diagrams showing Myopia and its correction.

8. Write the required material for the experiment to show that the ratio $\frac{V}{I}$ is a constant for a conductor.

   Explain the procedure of the experiment.

9. Hydrogen reacts with Oxygen to produce water. What will be the mass of water produced if 100 grams of Hydrogen participated in the reaction? Calculate the nubmer of molecules of water produced in this reaction. [Atomic masses : H = 1u, O = 16 u]

19E(B)

PART - B

Time : 15 min.] [Marks : 10

Instructions :

i) Answer **all** the questions.

ii) Each question carries **1 mark.**

iii) In this section there are 4 options (A / B / C / D) to each question. Choose the appropriate answer and write the answer in the brackets given against the question.

iv) **Part - B** must be attached to the answer booklet of **Part - A.**

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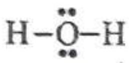
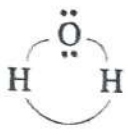
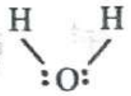
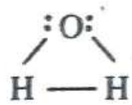
1. Identify the correct shape of water molecule [ ]

A)  B)  C)  D) 

2. Number of sub - shells in the main shell 'M' ______ [ ]

A) 1 B) 2 C) 3 D) 4

3. Which of the follwing gives the magnification for mirrors? [ ]

A) $-\frac{u}{v}$ B) $\frac{v}{u}$ C) $\frac{u}{v}$ D) $\frac{v}{u}$

4. Name of the Carbon Compounds $CH_3-CH_2-\underset{\substack{| \\ OH}}{C}=O$ [ ]

A) Ethanoic acid B) Propanoic acid C) Acetic acid D) Propanone

5. Which of the following element has more Atomic radius? [ ]

A) Li B) Na C) K D) Rb

6. Colour of Copper sulphate after removing water of crystallisation [ ]

A) White B) Blue C) Green D) Red

7. A double conves lens has two surfaces of equal radii 20 cm. Then the focal length is [ ]

A) 10 cm B) 20 cm

C) 40 cm D) Depends on refractive index

8. Example for polar covalent bond [ ]

A) $BeCl_2$ B) BF_3 C) HCl D) CH_4

9. Least energy orbital of the following [ ]

A) 4s B) 4p C) 4d D) 3d

10. Characteristics of the image formed by a convex mirror [ ]

i) Diminished ii) Enlarged iii) Inverted iv) Erected

A) i, iii B) i, iv C) ii, iii D) iii, iv